

Chapter 12: Introduction to Acid and Base

Textbook Exercise Solutions & Answers



1. Identify the following solutions, whether they are acid or base:

Solution	Change in Indicator	Acid / Base
1	No change (Phenolphthalein / Red litmus)	Acid
2	Orange colour turns red (Methyl orange)	Acid
3	Red litmus turns blue	Base

Reasons:

- **Solution 1 & 2 (Acid):** In acidic solutions, phenolphthalein remains colourless (no change), and methyl orange changes its orange colour to red/pink.
- **Solution 3 (Base):** In basic (alkaline) solutions, red litmus paper turns blue.

2. Write chemical names from given formulae. H_2SO_4 , $\text{Ca}(\text{OH})_2$, HCl , NaOH , KOH , NH_4OH

Write the chemical names from given formulae: H_2SO_4 , $\text{Ca}(\text{OH})_2$, HCl , NaOH , KOH , NH_4OH

1. H_2SO_4 – Sulphuric acid
2. $\text{Ca}(\text{OH})_2$ – Calcium hydroxide
3. HCl – Hydrochloric acid
4. NaOH – Sodium hydroxide
5. KOH – Potassium hydroxide
6. NH_4OH – Ammonium hydroxide

3. Sulphuric acid has highest importance in chemical Industry. Why?

Sulphuric acid has highest importance in the chemical industry because it is used in the manufacturing of fertilizers like ammonium sulphate and superphosphate of lime. It is used in the manufacturing of hydrochloric acid, nitric acid, phosphoric acid, ether, plastics and metal sulphates. It is also used in the manufacturing of dyes, drugs, perfumes, disinfectants and glue.

4. Give answers.

(a) Which acid is used for getting chloride salt?

Hydrochloric acid is used for getting chloride salt.

(b) By squeezing lemon on a piece of rock the gas liberated turned lime water milky. Which compound is present in the rock?

Metal carbonate is present in the rock.

(c) The label on the bottle of chemical is spoiled. How will you find whether the chemical is acidic or not?

The chemical is tested with blue litmus paper. If it turns red it is an acidic otherwise tested with red litmus paper, if it turns blue, it is an alkaline.

5. Answer the following questions.

(a) Explain the difference between acid and base.

Acids:

1. Acids have sour taste.
2. Acids turn blue litmus red.

Bases:

1. Bases have bitter taste.
2. Bases turn red litmus blue.

(b) Why indicator does not affect by salt?

Mostly indicators are organic compounds which do not react with a salt.

(c) Which substances are produced i by neutralization process?

Salt and water are produced in the f neutralization process.

(d) Which are the industrial uses of acids?

Acids which are used in the industry are:

1. Sulphuric acid
2. Hydrochloric acid
3. Nitric acid;

1. Sulphuric acid: Sulphuric acid is 5 used (a) in the manufacture of chemical? fertilizers like ammonium sulphate, (b) for cleaning gold and silver articles and (c) in car batteries.

2. Hydrochloric acid: Hydrochloric acid is used (a) to clean toilets and (b) to obtain glucose from starch and also for producing gelatine.

3. Nitric acid: Nitric acid is used (a) in the manufacture of perfumes (b) for engraving on copper, brass or silver and (c) in the manufacture of paints and explosives.

6. Select proper word given in bracket and fill in the blanks.

(a) Main constituent of acid is

Main constituent of acid is H⁺ ion.

(b) Main constituent of alkali is

Main constituent of alkali is OH⁻ ion.

(c) Tartaric acid is a acid.

Tartaric acid is a weak acid.

7. Match the pairs.

Group A	Group B
1. Tamarind	c. Tartaric acid
2. Curd	d. Lactic acid
3. Lemon	b. Citric acid
4. Vinegar	a. Acetic acid

8. State true or false.

(a) Oxides of metals are alkaline in nature.

True.

(b) Salt is acidic.

False. (Salt is neutral)

(c) Metal corrodes due to salts.

False. (Acids and bases corrode metals)

(d) Salts are neutral.

True. (Not all salts neutral)

9. Classify the following substances into acidic, basic and neutral group

HCl, NaCl, MgO, KCl, CaO, H₂SO₄, HNO₃, H₂O and Na₂CO₃.

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Group	Substances
Acid	HCl, H ₂ SO ₄ , HNO ₃
Base	CaO, MgO, Na ₂ CO ₃
Neutral	H ₂ O, NaCl, KCl.